Quiz3 CS221-2021 -30 Minutes

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Points: 34/45

1

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✓ **Correct** 1/1 Points

4

To specify behaviour of a 16 bit up/down counter using Moore FSM, how many number of states required and how many number of transition required?

- ◉ 2^16, 2^17 ✓
- ○ 2^16, 2^15
- ○ 16, 2^15
- ○ 15, 16

✗ **Incorrect** 0/2 Points

5

In designing an 8KB of 8-bit word size memory, how many D FFs required to design the memory including the memory storage, memory address register (MAR) and memory data register (MDR).

2064

Correct answers: 8210

✓ **Correct** 2/2 Points

6

Given one 3-bit synchronous counter and one 3-bit ripple counter made using flip flops having a propagation delay of 8 ns each. What will be the worst case delay in synchronous counter and ripple counter respectively

- ○ 24, 8
- ○ 8 ,8

- ◉ 8,24 ✓
- ○ 24,24

✓ **Correct** 1/1 Points

7

Given a D-FF, we want to design a JK-Flip Flop, What should the input of D FF in term of J, K and Q

- ○ D=K'Q'+JQ
- ○ D=KQ+J'Q'
- ◉ D=K'Q+JQ' ✓
- ○ D=KQ'+J'Q

✓ **Correct** 1/1 Points

8

Which of the following is true?

- ○ In sequential circuit output depends only on current input, but doesn't depend on past sequence.
- ◉ In sequential circuit output depends on both current input and on past sequence ✓
- ○ In combinational circuit output depends only on past sequence, but doesn't depend on current input.
- ○ None of the above

✓ **Correct** 1/1 Points

9

Which of the statements are false: Multiple answers

- [x] Synchronous counter and Asynchronous counter performs equally faster operations ✓
- [x] Synchronous counter is slower as compared to Asynchronous counter ✓
- [ ] Ripple carry counter is a Asynchronous counter
- [ ] Synchronous counter is faster as compared to Asynchronous counter

✗ **Incorrect** 0/2 Points

10

Given a Moore FSM with 8 states and two one-digit decimal input and 2 binary output. Calculate the number of possible transitions in transition function.

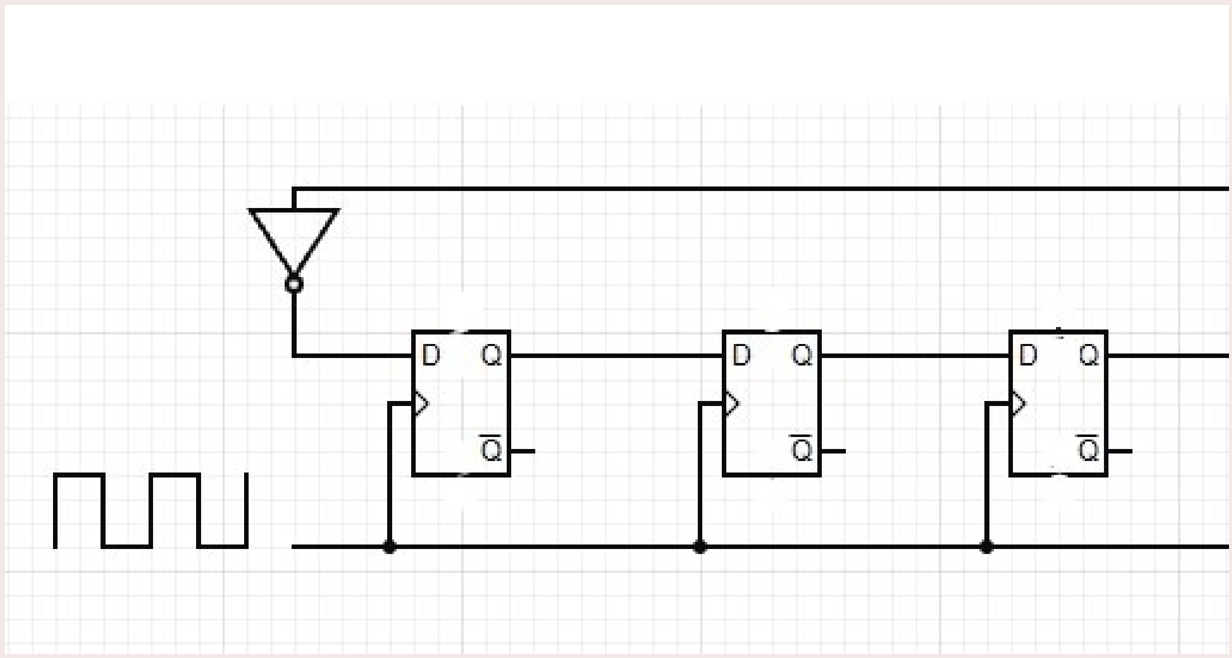
56

Correct answers: 800

✓ **Correct** 2/2 Points

11

Given 4-bit register initially set to 1001. After how many clock cycles the register will set to the same sequence 1001



- (selected) 8 ✓
- 7
- 4
- 5

✓ **Correct** 2/2 Points

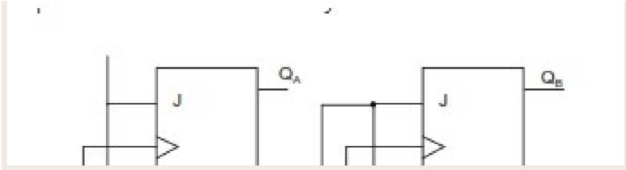
12

A RS latch can be constructed using

- [x] Two cross coupled NAND gates ✓
- [ ] Two cross coupled OR gates
- [ ] Two cross coupled AND gates
- [ ] Two cross coupled XOR gates ✓

✗ **Incorrect** 0/2 Points

13



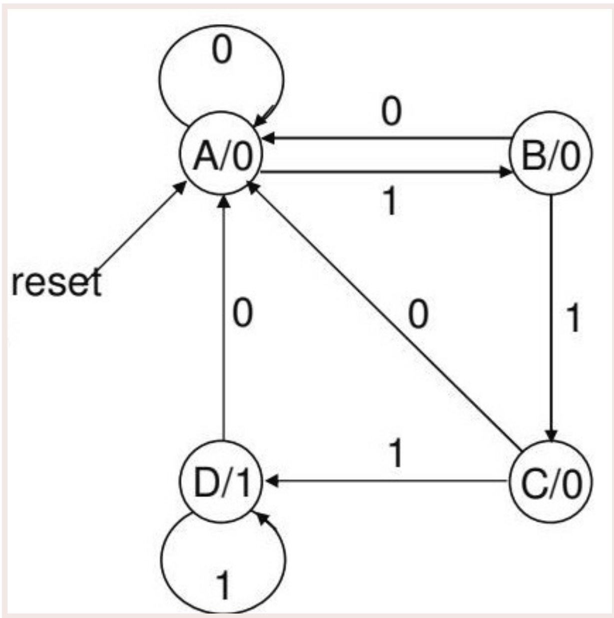
Given the figure, If the input sequence is 1010 from left to right and the initial value of Qa and Qb of two J-K flip flop is 00 then find the output sequence of Qb.

- 0101
- 0010 ✓
- 1011
- non of above

✓ **Correct** 2/2 Points

14

Assume the following Moore state machine for a string of 0s and 1s. This can be used for detect



- Number of 1's more than 3
- At least one occurrence of 3 or more continuous 1's ✓
- At least one occurrence of 4 continuous 1's
- Number of 0s multiple of 3

✓ **Correct** 1/1 Points

15

A new flip flop (named as XY Flip Flop) is designed with having the following behaviour as described. It has two inputs X and Y and when both inputs are same and they are X=1, Y=1 the flip flop is going to set and when they are X=0, Y=0 flip flop going to reset, if both inputs are different and X=0,Y=1 the flip flop complements itself otherwise if X=1,Y=0 the flip flop going to retain the last state. which of the following expression is the characteristic expression for new flip flop?

- ○ Q(N+1)=YQ+XQ'
- ○ Q(N+1)=XQ'+YQ'
- ◉ Q(N+1)= XQ+YQ' ✓
- ○ All of the above

✓ **Correct** 2/2 Points

16

The minimum number of filp flop required for a counter which counts the sequence of numbers :12, 1024, 8, 5, 6, 7, 8, 9 and 128?

4

✓ **Correct** 2/2 Points

17

We want to design a configurable logic block (CLB) with 8 bit PIPO and Multiplexor to implement any Boolean function of three input. Boolean function of the can be defined by contents of 8 bit PIPO in truth table storage format. What will be required size of the Multiplexor (______to 1) to implement the same? Fill the blank

8

Incorrect 0/1 Points

18

Given a FSM with two input A and B, and two state S0 and S1, the state changes happens when AB=10 and AB=01, and rest of the case it stays in the same state. Calculate the number of 3 input AND gate and 2 input OR gate required to implement the same. Ignore the number of not GATE.

- 4, 3 ✓
- 3, 2
- 4, 2
- 1, 4

Correct 1/1 Points

19

Which of the following statement is true?

- Flipflops are level sensitive whereas latches are edge sensitives.
- Flipflops are edge sensitive whereas latches are level sensitives. ✓
- Both flipflops and latches are edge sensitive.
- Both flipflops and latches are level sensitive

Correct 1/1 Points

20

Minimum number of flip-flops required to design a binary counter which count from 0 to 64 and repeat.

- 65
- 7 ✓
- 6
- 32

✓ **Correct** 1/1 Points

21

What is the minimum number of NAND gate required for positive edge triggered D-flip flop

- 6 ✓
- 4
- 8
- 3

✓ **Correct** 1/1 Points

22

Output of FSM needs to be always same as state-encoding of the FSM

- True
- False ✓

Incorrect 0/2 Points

23

What is race condition in SR latch with two cross coupled XOR gates? Choose all the true statements

- [ ] When SR=11, both Q and Q' are set to 0 ✓
- [x] When SR=11, both Q and Q' are set to 1
- [ ] When SR=00, after SR=11, it is difficult predict the value of Q' ✓
- [x] When SR=00, after SR=11, it is difficult predict the value of Q ✓

Correct 2/2 Points

24

What will be the sum of parameters (number of states, number of inputs, total number of transitions and total number of outputs) in the complete Moore Machine specification for JK flip flop will be

14

Correct 2/2 Points

25

Suppose we want to design a multifunction register with supporting six data modification (including retaining older one) function, what will be the size of the multiplexor in term of number of select line to be used in the design?

3

✓ **Correct** 2/2 Points

26

In a FSM implementation which one is correct?

- ○ Propagation delay of Next state logic should be higher than clock cycle
- ◉ Propagation delay of Next state logic should be less than clock cycle ✓
- ○ Propagation delay of Next state logic should be exactly equal to the clock cycle
- ○ No relation

✓ **Correct** 1/1 Points

27

How much time is required to transfer a word using serial transfer using shift register if word size is 8 bit and frequency of clock is 50KHz.

- ○ 1.6 x10^(-3)s
- ◉ 0.16x 10^(-3)s ✓
- ○ 0.14 x 10^(-3)s
- ○ 1.4x 10^(-3)s

✓ **Correct** 1/1 Points

28

Which one is Universal flip-flop

- ○ SR
- ◉ JK ✓
- ○ D
- ○ T

✓ **Correct** 1/1 Points

29

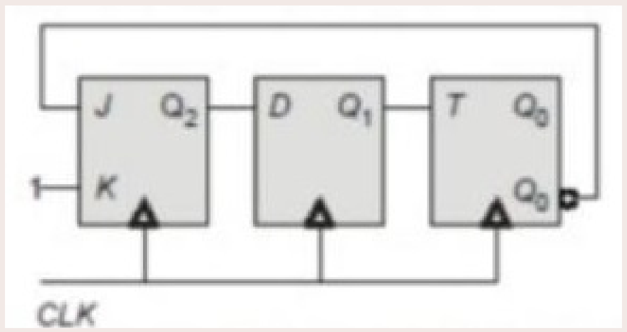
Main reason to use Master slave configuration of two latches to form a flip flop is

- Stabilize the input SR to store properly
- Ensure no race condition occurs in RS latch
- Convert level sensitive to edge sensitive ✓
- Convert edge sensitive to level sensitive

✗ **Incorrect** 0/2 Points

30

For given counter with JK, D and T FFs, the initial state is 101 for Q2Q1Q0. After 4 clock pulses state (Q2 Q1 Q0) of the counter will be



- 101
- 011
- 010 ✓
- 100

✓ **Correct** 1/1 Points

31

Race condition occur in J-K latch when-

- ○ Enable=0, J=1, K=1
- ○ Enable=1, J=0, K=0
- ◉ Enable=1, J=1, K=1 ✓
- ○ Enable=0, J=0, K=0

✓ **Correct** 1/1 Points

32

Which of the following is characteristic equation of SR flip-flop

- ○ Q(n+1) = S + R Q(n)′
- ○ Q(n+1) = S′ + R Q(n)
- ○ Q(n+1) = S + R′ Q(n)′
- ◉ Q(n+1) = S + R′ Q(n) ✓

✓ **Correct** 2/2 Points

33

We want to implement FSM with negative edge triggered (NET) FF instead of positive edge triggered (PET) FF, what should we do?

- ○ FSM implementation will not work with NET FFs
- ○ Need to convert the NET FF to PET FF
- ○ We need to redesign the FSM implementation
- ◉ Same FSM implementation of PET FF works fine with NET FF ✓

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